

# http-api

## HTTP API Contract — Workshop Curriculum Platform

**Feature:** specs/001-workshop-curriculum-platform | **Phase:** 1 (Design & Contracts) | **Date:** 2026-09-01

**Governs:** the backend HTTP surface consumed by the Angular client.  
**Derived from:** [spec.md](#) (FR/SC), [plan.md](#) (five corrected premises), [research.md](#) (D-SEARCH-\*, D-LLM-\*), [data-model.md](#). **Companion contracts:** [passage-contract.md](#) (what a pid is), [pipeline-cli.md](#) (how the data this API serves is produced).

This document specifies **interfaces and behavioural guarantees**. It contains no handler bodies, no business logic and no migrations. Where a guarantee cannot be offered because something is unverified, it is marked and the settling measurement is named (§8) rather than assumed away.

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## 1. Conventions

### 1.1 Base, transport, scope

| Property     | Value               | Basis                                                                                   |
|--------------|---------------------|-----------------------------------------------------------------------------------------|
| Base path    | /api                | FR-013 — matches<br>ai_interviewing/<br>platform/backend/<br>internal/api/<br>api.go:36 |
| Bind address | loopback by default | D1 (local/internal<br>only)                                                             |

| Property          | Value                                                                                                           | Basis                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Authentication    | <b>none</b>                                                                                                     | D1 — single machine, single user; no accounts are in the data model |
| Request encoding  | application/json for bodies; query string for reads                                                             |                                                                     |
| Response encoding | application/json; charset=utf-8, except recording/material bytes                                                |                                                                     |
| Versioning        | none in the path. The contract is versioned with the repository. Breaking changes bump X-Workshop-Api-Contract. |                                                                     |

There is **no** rate limiting, quota, or tenancy. Adding any would imply a multi-user deployment, which D1 excludes.

## 1.2 Response headers required on every /api response

| Header                  | Meaning                                                                                        |
|-------------------------|------------------------------------------------------------------------------------------------|
| X-Workshop-Api-Contract | Contract revision this build implements (e.g. 1).                                              |
| Cache-Control           | no-store on every three-state endpoint (\$2). Only recording and material bytes are cacheable. |

Additionally, on every endpoint that reads the index or the passage registry:

| Header                      | Meaning                                                                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| X-Workshop-Index-Generation | Integer generation the response was served from (data-model Index.generation). Lets the client detect an atomic swap mid-session and re-fetch. |
| X-Workshop-Search-Status    |                                                                                                                                                |

| Header | Meaning                                                                                                                                        |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------|
|        | Mirrors the body's status verbatim. <b>Convenience mirror only</b> . It MUST equal <code>body.status</code> ; a gate asserts agreement (§2.7). |

X-Workshop-Search-Status exists so that a proxy, a log line or a curl -I can distinguish the three states without parsing a body. It is never the authority — the body is.

### 1.3 Why Cache-Control: no-store is mandatory on the three-state endpoints

A cached 200 ok outliving an index swap or a backend failure would let an intermediary serve a successful-looking response while the backend is unavailable. That is precisely the FR-020 violation this contract exists to make impossible, so caching is removed rather than tuned.

### 1.4 Path parameters

| Parameter | Form                                   | Notes                                                                                                                                                              |
|-----------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| {chapter} | chapter slug (data-model Chapter.slug) | Immutable once published. The zero-padded ordinal is <b>not</b> accepted as a path key — one key, one meaning.<br><br>See <a href="#">passage-contract.md §2</a> . |
| {pid}     | canonical uppercase ULID, 26 chars     | Lowercase input is accepted and canonicalised; responses always carry uppercase.                                                                                   |
| {job_id}  | opaque string                          | Answer job handle. Not a ULID; carries no meaning.                                                                                                                 |

## 1.5 Malformed-request errors (4xx) are a fourth thing, not a fourth state

A 4xx says *the request was wrong*. It is outside the three-state contract of §2, which describes outcomes of well-formed requests. The client must not render a 4xx through any of the three result branches.

```
// 4xx body – the only shape a 4xx ever has
{
  "error": {
    "code": "empty_query",           // closed enum, see §5.2
    "message": "q is required and must be non-empty after
trimming",
    "field": "q"                     // optional
  }
}
```

---

## 2. The three-state contract (FR-019, FR-020, FR-033, SC-013, D-SEARCH-4)

This is the load-bearing section of this document. Everything else is plumbing.

### 2.1 The requirement, restated exactly

There are **three** distinguishable outcomes, never two:

| State       | Meaning                                                              | Constitution analogue        |
|-------------|----------------------------------------------------------------------|------------------------------|
| ok          | The question was answered and there is something to show.            | exit 0                       |
| no_match    | The question was answered and the answer is <i>nothing matches</i> . | exit 1 — checked, negative   |
| unavailable | <b>The question could not be answered.</b>                           | exit 2 — COULD NOT DETERMINE |

The `.specify/memory/constitution.md` **Honest Instruments** principle is binding here: collapsing state 2 into state 0 makes a broken system certify content nobody searched; collapsing it into state 1 makes a broken backend assert that the curriculum contains nothing. Both are release blockers.

## 2.2 The anti-pattern this repository already ships

`ai_interviewing/platform/frontend/src/app/features/search.component.ts` is the concrete failure the contract is designed against. Measured, at the lines quoted:

```
// line 19 – the only negative branch in the template
@else if (q().length >= 2 && !hits().length) { <p
    class="muted">No results for "{{ q() }}"</p> }

// line 58 – the error branch sets no error state at all
).subscribe({ next: h => { this.hits.set(h || []);
    this.loading.set(false); },
    error: () => this.loading.set(false) });
```

A backend failure clears `loading`, leaves `hits` empty, and the template renders “**No results for ...**”. The component has two states where three are required. Its backend compounds this by returning `[]` for queries shorter than two characters. This defect exists independently of this feature and must not be inherited.

## 2.3 Why liveness cannot gate the third state

Measured in one MCP session, seconds apart (`research.md` D-SEARCH-4):

|                             |                                |
|-----------------------------|--------------------------------|
| == health_check 2 ms ==     | == semantic_search 20013 ms == |
| Status: OK                  | embed query: all embedding     |
| servers exhausted after     |                                |
| Message: service is healthy | failover: ... context deadline |
| exceeded [isError=True]     |                                |

`health_check` reported healthy **2 ms before** the search path failed hard. It is a liveness probe, not a saturation probe.

**Normative:** the `status` field **MUST** be derived from the outcome of the search call actually performed for this request. It **MUST NOT** be derived, in whole or in part, from `/api/health`, from Lumen’s

health\_check, from a cached liveness flag, or from any probe that did not execute this query. /api/health (§3.1) exists for process supervision only and carries an explicit non-guarantee in its own response.

## 2.4 Why upstream text is quarantined

Lumen returns, as **one unstructured string** (research.md D-SEARCH-4):

```
"No results found. | Warning: Index is being updated in the
background.
```

```
Results may be incomplete or outdated. Us..."
```

A client that reads the first line, or renders the blob, displays “No results found.” while the index is provably incomplete.

### Normative:

- The backend **MUST** parse upstream result text for degradation markers, and **MUST** promote a match to degraded.semantic = "reindexing".
- Upstream free text **MUST NEVER** appear in results[], in suggestions[], or in any field the client renders as primary copy.
- Upstream text may appear **only** in reason.evidence or degraded.evidence, both of which are declared **diagnostic-only** fields. The client contract forbids rendering them outside a diagnostics disclosure.

## 2.5 The response envelope

Returned by /api/search, /api/suggest, and /api/passages/{pid}/crossrefs, with the payload array named per endpoint.

```
{
  "status": "ok",                      // "ok" | "no_match" |
  "unavailable" – exhaustive, closed
  "query": "how do we cut chunks at silence",
  "generation": 7,                     // index generation; null iff
status=="unavailable"
  "floor_calibrated": false,           // §8 U5 – see 2.6 rule I7
  "legs": {                            // per-leg outcome; keys
present iff the leg was enabled
    "lexical": "ok",                  // "ok" | "failed" | "skipped"
    "semantic": "failed"
  },
  "results": [ /* Hit – §3.7 */ ], // PRESENT AND NON-EMPTY IFF
status=="ok". ABSENT otherwise.
  "near_misses": [ /* Hit */ ],      // present only when
```

```

status=="no_match"; see I6
  "degraded": {                                // present only alongside
status=="ok". Never with the others.
    "semantic": "reindexing",                  // closed enum, §5.3
    "message": "Meaning-based search is unavailable right now –
showing keyword matches only.",
    "evidence": "No results found. | Warning: Index is being
updated..." // DIAGNOSTIC ONLY
  },
  "reason": null,                               // present IFF
status=="unavailable"; shape in §2.6
  "took_ms": 41
}

```

unavailable bodies additionally carry:

```

{
  "status": "unavailable",
  "reason": {
    "code": "embedding_backend_exhausted", // closed enum, §5.3
    "leg": "semantic",                     // "lexical" |
"semantic" | "index" | "registry"
    "message": "Search cannot answer right now: the embedding
backend is saturated.",
    "retry_after_s": 30,                   // nullable – null
means UNKNOWN, never 0
    "evidence": "all embedding servers exhausted after failover: ...
context deadline exceeded"
  }
}

```

## 2.6 Invariants — these are what make the anti-pattern physically impossible

| #  | Invariant                                                                                               | Consequence for a naive client                                                                         |
|----|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| I1 | results is present <b>iff</b> status == "ok", and when present it is <b>non-empty</b> . It is never []. | hits.length == 0 can no longer mean “no results” — on any non-ok outcome there is no array to measure. |
| I2 | status == "unavailable" <b>iff</b> HTTP 503. Never 200.                                                 | A client that only branches on HTTP status still cannot render “nothing                                |

| #  | Invariant                                                                                                                                                                                                                                                                           | Consequence for a naive client                                                                                                                                                    |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I3 | <p><code>status == "no_match" ⇒ HTTP 200.</code> <code>status == "ok" ⇒ HTTP 200.</code></p>                                                                                                                                                                                        | <p>found” for a backend failure.</p> <p>Determined negatives are successes, so <code>no_match</code> stays measurable (it is a real result, not an error).</p>                    |
| I4 | <p>reason is present <b>iff</b> <code>status == "unavailable"</code>, and its code is from the closed enum in §5.3.</p>                                                                                                                                                             | <p>No free-form failure strings reach the UI.</p>                                                                                                                                 |
| I5 | <p><b><code>no_match</code> requires every <i>enabled</i> leg to have returned successfully.</b> If any leg failed <b>and</b> the surviving legs produced zero results, the outcome is <code>unavailable</code> with <code>reason.code = "partial_failure_zero_results"</code>.</p> | <p>The single most-missed case: a degraded search may report ok; it may <b>never</b> report <code>no_match</code>. “The leg that ran found nothing” is not “nothing matches”.</p> |
| I6 | <p><code>near_misses</code> may only appear with <code>no_match</code>, every entry carries <code>"above_floor": false</code>, and the client contract requires them to be rendered in a visually distinct, explicitly labelled region.</p>                                         | <p>FR-019 — below-threshold items are shown as context, never as answers.</p>                                                                                                     |
| I7 | <p>When <code>floor_calibrated == false</code>, a <code>no_match</code> <b>MUST</b> be rendered with the calibration caveat, because an uncalibrated relevance floor cannot support the claim “nothing matches”.</p>                                                                | <p>§8 U5. Honest until measured.</p>                                                                                                                                              |
| I8 | <p><code>degraded</code> may accompany <code>ok</code> only. It is never present with <code>no_match</code> (excluded by I5) nor with <code>unavailable</code> (which is total, not partial).</p>                                                                                   | <p>One banner, one meaning.</p>                                                                                                                                                   |

| #  | Invariant                                                                                                                 | Consequence for a naive client                                      |
|----|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| I9 | The envelope is constructed in exactly <b>one</b> function in the search service. No handler assembles it field-by-field. | Makes I1–I8 enforceable at a single point rather than per-endpoint. |

## 2.7 Gates and their paired mutations (FR-032, SC-012)

Every guarantee above is a claim, so each carries a gate and a mutation proving the gate fails when the guarantee is broken. Constitution §1.1: *a gate that has never been observed failing is not known to work.*

| Gate     | Assertion (executes the behaviour, does not grep for it)                                                                                                                                         | Paired mutation that MUST turn it red                                                                                                          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| G-HTTP-1 | Point the embedding endpoint at a closed port; GET /api/search?q=x returns <b>503</b> with status:"unavailable" and a reason.code.                                                               | Make the handler return 200 {"results":[]} on backend error. Gate must FAIL.                                                                   |
| G-HTTP-2 | Same failure; assert the response body has <b>no</b> results key at all.                                                                                                                         | Add "results": [] to the unavailable branch. Gate must FAIL.                                                                                   |
| G-HTTP-3 | Fail the semantic leg, let lexical return zero rows; assert status == "unavailable", reason.code == "partial_failure_zero_results".                                                              | Change the rule to emit no_match when any leg succeeded. Gate must FAIL. (This is invariant I5, the one most likely to be “simplified” later.) |
| G-HTTP-4 | Feed the search service Lumen’s literal "No results found. \Warning: Index is being updated..." string; assert status=="ok" with degraded.semantic=="reindexing" when lexical has rows, and that | Forward the upstream string into results[0].snippet. Gate must FAIL.                                                                           |

| Gate     | Assertion (executes the behaviour, does not grep for it)                                                                                                                                   | Paired mutation that MUST turn it red                                                                                              |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
|          | the raw string appears <b>only</b> under degraded.evidence.                                                                                                                                |                                                                                                                                    |
| G-HTTP-5 | With ollama stopped: GET /api/search returns <b>200</b> with real lexical results <b>and</b> POST /api/ask returns <b>503</b> state:"unavailable".                                         | Wire /api/search through the answering provider's health check. Gate must FAIL. (FR-025; mirrors research llm-bridging §3 item 5.) |
| G-HTTP-6 | For 100 randomised outcomes, assert X-Workshop-Search-Status header equals body.status.                                                                                                    | Hardcode the header to ok. Gate must FAIL.                                                                                         |
| G-HTTP-7 | Frontend: render each of the three envelopes; assert the no-results copy does <b>not</b> appear for the unavailable envelope and the unavailable copy does <b>not</b> appear for no_match. | Restore the ai_interviewing error handler (error: () => loading.set(false)). Gate must FAIL.                                       |
| G-HTTP-8 | Packet/socket assertion: during 200 /api/suggest calls, zero requests reach the embedding endpoint.                                                                                        | Add an embedding call to the suggest path. Gate must FAIL. (§3.6, SC-005.)                                                         |

Evidence for every gate is written to the versioned evidence location per FR-040 — see [pipeline-cli.md §2.5](#).

### 3. Endpoint catalogue

Legend: **Traces** names the requirement each endpoint exists to satisfy. An endpoint with no trace would be unrequested capability and is not in this contract.

### 3.1 GET /api/health — process liveness only

**Traces:** none (operational). Present because ai\_interviewing has it (FR-013) and the control plane's status.sh needs it ([pipeline-cli.md §4.8](#)).

#### Response 200

```
{
  "ok": true,
  "version": "...",
  "contract": 1,
  "index_generation": 7,
  "not_a_search_availability_signal": true
}
```

The last field is not decoration. It is the contract's explicit non-guarantee, carried in-band so that a reader of a captured response cannot mistake this endpoint's meaning:

**This endpoint reports that the HTTP process is up. It says nothing about whether search can answer.** Using it to gate search availability reproduces the measured health\_check failure in §2.3. Search availability is only ever the status field of a search response.

/api/health never returns 503 for a degraded backend; if the process can answer, it answers 200.

---

### 3.2 GET /api/chapters — list chapters

**Traces:** FR-008, FR-013, SC-004 (the first screen a new user reaches).

**Request:** no parameters. (Scale is “one chapter, growing to tens” — spec Assumptions — so no pagination is specified. Adding pagination would be unrequested capability.)

#### Response 200

```
{
  "chapters": [
    {
      "slug": "01-ai-workflows",
      "ordinal": 1,
      "title": "Chapter 1 – Session Recording",
      "summary": "...",
      "status": "published",           // "draft" | "transcribed"
    }
  ]
}
```

```

| "published"
  "recording": {
    "available": true,           // false ⇒ parts present
    "duration_s": 6928.75,      // measured, from the
Recording entity
    "size_bytes": 1871981557
  },
  "transcript": {
    "present": true,
    "passage_count": 1421,
    "uncertain_count": 37,      // FR-003 – visible before
opening the chapter
    "coverage_complete": true,  // SC-001 arithmetic
identity, computed at ingest
    "accuracy": { "measured": false, "wer": null } //
FR-004 / SC-002; null until measured
  },
  "material_count": 1
}
],
"generation": 7
}

```

accuracy.wer is null, never a placeholder number, until verify-accuracy.sh has run. measured: false with wer: null is the “unable to verify” state (FR-033) expressed in data.

**Errors:** 503 {"status": "unavailable", "reason": {"code": "registry\_unreadable"}} if the passage registry cannot be opened. Never an empty chapters array for a storage fault — that is I1 applied to browsing.

---

### 3.3 GET /api/chapters/{chapter} — one chapter with its materials

**Traces:** FR-008, FR-009, FR-011.

#### Response 200

```

{
  "slug": "01-ai-workflows",
  "ordinal": 1,
  "title": "...",
  "summary": "...",
  "status": "published",
  "sections": [                // FR-011 – navigable,
never one undifferentiated block
    { "id": "s-03", "title": "Cutting chunks at silence",

```

```

"first_pid": "01JBX...", "t_start_s": 512.4 }
],
"materials": [
  {
    "id": "notes-gemini",
    "kind": "pdf", // "pdf" | "markdown" |
    "image" | "diagram" | "code"
    "title": "Notes by Gemini",
    "bytes": 421888,
    "href": "/api/chapters/01-ai-workflows/materials/notes-
gemini",
    "authority": "secondary" // D-TRANS-5 – the
recording is authoritative
  }
],
"recording": { "href": "/api/chapters/01-ai-workflows/
recording",
                "probe": "/api/chapters/01-ai-workflows/
recording/probe" },
"transcript": { "href": "/api/chapters/01-ai-workflows/
transcript" },
"accuracy": { "href": "/api/chapters/01-ai-workflows/
accuracy" }
}

```

authority: "secondary" is contractual, not cosmetic: the notes PDF is a ~26-words-per-minute summary that demonstrably errs (“Spatkit” for SpecKit). The client must present it as supporting material, never as the record of what was said (spec edge case: *“The notes PDF and the transcript disagree — the recording is the authority”*).

**Errors:** 404 chapter\_not\_found (determined negative); 503 registry\_unreadable.

---

### 3.4 GET /api/chapters/{chapter}/transcript — passages, timestamps, provenance, uncertainty

**Traces:** FR-001, FR-002, FR-003, FR-005, FR-011, FR-038, SC-001, SC-003.

#### Request

| Parameter | Type | Default | Notes                                 |
|-----------|------|---------|---------------------------------------|
| from_pid  | ULID | —       | Window start; omit for the beginning. |
| limit     | int  | 500     | Max 2000. A 1,400-passage             |

| Parameter                | Type   | Default | Notes                                                                          |
|--------------------------|--------|---------|--------------------------------------------------------------------------------|
|                          |        |         | chapter is servable whole; the window exists so a client need not hold it all. |
| section                  | string | —       | Restrict to one sections[].id.                                                 |
| include_redacted_markers | bool   | true    | Redacted passages appear as markers, never as content (FR-039).                |

**Response 200**

```
{
  "chapter_slug": "01-ai-workflows",
  "engine": "faster-whisper/large-v3-turbo",          //
Transcript.engine
  "engine_params": { "beam_size": 5, "condition_on_previous_text":
false,
                    "normaliser_sha256": "sha256:..." },    //
frozen before first measurement
  "duration_s": 6928.75,
  "coverage": {                                           // SC-001 as
an arithmetic identity, D-TRANS-4
    "complete": true,
    "speech_span_s": 6887.42,
    "accounted_silence_s": 41.33,
    "unexplained_gap_s": 0.0,                           // MUST be
exactly 0 for a published chapter
    "gaps": []                                           // each gap:
{start_s, end_s, reason}
  },
  "passages": [
    {
      "pid": "01JBX7QK3M8V2ZC4YT5N6RWDPA",
      "kind": "transcript",
      "text": "we always cut inside a measured silence, never mid-
word",
      "t_start_s": 512.4,
      "t_end_s": 519.1,
```

```

        "speaker": "Milos",                // null when not human-
attributed – never machine-guessed
        "speaker_source": "human",         // "human" |
"unattributed" – D-TRANS-2, dual-mono audio
        "provenance": "human_corrected",   // "machine" |
"human_corrected" (FR-038)
        "confidence": -0.31,              // avg_logprob-derived;
null for non-transcript kinds
        "uncertain": false,                // FR-003
        "uncertain_reason": null,          // "low_confidence" |
"inaudible" | "overlapping_speech"
                                           // |
"language_uncertain" – null iff uncertain==false
        "redacted": false,
        "content_hash": "sha256:...",     // CHANGE DETECTION ONLY
– never identity. See passage-contract §6.
        "machine_text": "we always cut inside a measured silence
never mid word"
                                           // present iff
provenance=="human_corrected"; the
                                           // immutable recogniser
output, retained as evidence
    },
    {
        "pid": "01JBX7QK55T0N9E1XJ2H3KFMQR",
        "kind": "transcript",
        "text": null,                      // redacted passages
carry no text, at any layer
        "t_start_s": 1204.0, "t_end_s": 1211.6,
        "redacted": true,
        "redaction": { "reason_code": "third_party_pii", "at":
"2026-09-01T10:02:11Z" }
    }
],
    "next_from_pid": "01JBX7QK9..."     // null at the end of the
chapter
}

```

## Contractual points

- speaker is null unless a human attributed it. The recording’s “stereo” is **dual-mono** (L-R = -90.3 dB) and uniformly AGC-compressed, so channel- and loudness-based separation are measurably impossible (D-TRANS-2). FR-005 explicitly permits stating inability; the API states it through speaker\_source: “unattributed” rather than by guessing.
- uncertain is set from the engine’s own confidence signal, never inferred downstream. A passage with uncertain: true MUST carry a non-null uncertain\_reason.

- `machine_text` is served alongside corrected text so a reader can see both layers. Deleting it would destroy the ability to re-measure accuracy later (SC-002); the API therefore exposes it rather than only the current text.
- `coverage.unexplained_gap_s > 0` on a chapter with status: "published" is a contract violation; the publish step refuses it ([pipeline-cli.md §4.7](#)).

**Errors:** 404 `chapter_not_found`; 404 `transcript_not_produced` (determined: the chapter exists and has no transcript — distinct from the chapter not existing); 503 `registry_unreadable`.

---

## 3.5 Recording: serve and seek a local file (D1, D3)

There is no streaming service here. The 1.8 GB file is reassembled locally from  $36 \times 50$  MB parts, checksum-verified, and byte-served. **No transcoding, no HLS/DASH, no adaptive bitrate ladder, no CDN** — D3 excludes all of it.

### 3.5.1 GET `/api/chapters/{chapter}/recording/probe`

**Traces:** FR-007, FR-009, SC-003.

```
{
  "available": true,
  "duration_s": 6928.75,
  "size_bytes": 1871981557,
  "sha256": "...",
  "verified_at": "2026-09-01T09:12:44Z",
  "container": "mp4", "video": "h264 1920x1080 24fps", "audio":
  "aac-lc 48000 stereo(dual-mono)",
  "etag": "\"sha256-1f4c9ab0e7d2...-1871981557\""
}
```

When not available, probe returns **503**, not 200-with-available:false, for the same reason searches do (§2.6 I2):

```
{
  "status": "unavailable",
  "reason": {
    "code": "recording_not_reassembled",
    "message": "The recording has not been reassembled from its 36
archive parts.",
    "remedy": "bash workshop/scripts/extract-videos.sh",
    "parts_present": 36, "parts_expected": 36
  }
}
```

```

    }
}

```

### 3.5.2 GET|HEAD /api/chapters/{chapter}/recording

**Traces:** FR-009, FR-007, SC-003, D3.

#### Range support (mandatory)

| Aspect               | Contract                                                                                                                                                                                                                                                            |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accept-Ranges: bytes | Sent on GET and HEAD, always.                                                                                                                                                                                                                                       |
| Accepted forms       | bytes=N-, bytes=N-M, bytes=-N (suffix).                                                                                                                                                                                                                             |
| Multi-range          | <p><b>Not supported.</b></p> <p>bytes=0-99,200-299 ⇒ 416 with error.code = "multi_range_unsupported".</p> <p>Returning the full entity instead (also RFC-permitted) would push 1.8 GB down a link that asked for 200 bytes, so the contract refuses explicitly.</p> |
| Success              | 206 Partial Content with Content-Range: bytes a-b/total and Content-Length = part length.                                                                                                                                                                           |
| Unsatisfiable        | 416 Range Not Satisfiable with Content-Range: bytes */<total>.                                                                                                                                                                                                      |
| No Range header      | 200 with the full entity and Content-Length. Never chunked-with-206.                                                                                                                                                                                                |
| If-Range             | Honoured against the strong ETag: match ⇒ 206; mismatch ⇒ 200 full entity.                                                                                                                                                                                          |
| ETag                 | Strong, content-addressed: "sha256-<first 16 hex of the verified digest>-<size>". Stable across a re-extraction on another machine, because it is derived from the manifest digest, not from inode metadata.                                                        |
| Concurrency          | Bounded by WORKSHOP_RECORDING_MAX_READERS                                                                                                                                                                                                                           |

| Aspect | Contract                                                                                                  |
|--------|-----------------------------------------------------------------------------------------------------------|
|        | (derived from host CPU count, overridable). Exceeded $\Rightarrow$ 503 too_many_readers with Retry-After. |

### Integrity (FR-007) without hashing 1.8 GB per request

The extraction step writes a sidecar recording the verified sha256 together with the size and mtime of the file it verified. On each request the server compares the current file’s size and mtime against the sidecar:

| Condition                             | Response                                                            |
|---------------------------------------|---------------------------------------------------------------------|
| Sidecar matches the file              | Serve.<br>503                                                       |
| Sidecar absent                        | recording_integrity_unverified — bytes are never served unverified. |
| Sidecar present, size/mtime differ    | 503 recording_integrity_stale, remedy: re-run extraction.           |
| Extraction recorded a digest mismatch | 503 recording_integrity_mismatch — “fail loudly” (FR-007).          |

All four are **state 2**, not 404 and not a partial serve. The distinction matters: 404 on this route means *this chapter has no recording at all* (a determined negative); “the file has not been reassembled yet” is *could not determine*, and a fresh clone is in exactly that state.

### Seeking (SC-003)

The client seeks by setting `currentTime` on the media element from a passage’s `t_start_s`; the browser then issues its own Range requests using the MP4’s internal index. The API’s only obligation is honest range support.

**Normative non-capability:** the server **MUST NOT** expose a “seek to N seconds” endpoint. Mapping a timestamp to a byte offset requires an index the container already holds and the browser already uses; re-deriving it server-side would be invented capability with a new failure mode and no requirement behind it.

**Caching:** Cache-Control: private, max-age=31536000, immutable plus the strong ETag. This is the one route where caching is correct, because the entity is content-addressed and immutable (FR-006 — source material is never modified).

**3.5.3 GET /api/chapters/{chapter}/materials/{material\_id}**

**Traces:** FR-009, FR-016. Byte-serves a supporting material with its declared Content-Type. Same ETag/range rules as §3.5.2 (the 412 KB notes PDF does not need them, but one rule is fewer rules). 404 chapter\_not\_found when the slug names no chapter; 404 material\_not\_found when the chapter is real and holds no material under that id; 503 when the file is listed in the chapter manifest but missing on disk — again, absent-but-expected is state 2.

**That 503 carries reason.code = "curriculum\_unreadable" (§5.3), and every 503 this route can produce carries the same one.** The clause above used to specify the condition and name no code at all, which left the one field a client is contractually required to branch on (§6) unstated for a response this contract says is reachable — measured and named 2026-09-03, and the omission is **WITHDRAWN**. There are four such paths and they are one code because they are one remedy — the chapter tree could not be read — while reason.message and the diagnostic reason.evidence string distinguish them:

| Condition                                                            | reason.code           | reason.leg |
|----------------------------------------------------------------------|-----------------------|------------|
| The chapter tree could not be listed, so no slug can be resolved     | curriculum_unreadable | curriculum |
| The chapter's own directory could not be listed                      | curriculum_unreadable | curriculum |
| The material is listed in the chapter and could not be <b>opened</b> | curriculum_unreadable | curriculum |
| The material was opened and could not be <b>measured</b> (stat)      | curriculum_unreadable | curriculum |

Read from the four non-test raise sites in `workshop/platform/backend/internal/api/chapters.go` (cited **by path only**), each calling the single `writeUnavailable` helper in that same file. That helper is the only way a 503 leaves this handler, and it emits `status: "unavailable"` plus a reason object with `code`, `leg`, `message` and `retry_after_s`, together with the `X-Workshop-Search-Status: unavailable` header — the §4 shape exactly. `curriculum_unreadable` is an existing §5.3 member and no code was invented to fill this gap.

**Honest boundary (§11.4.6): this is read from source, not from a live probe.** All four conditions require the chapter tree to be unreadable underneath a running server, which was not induced on the live deployment. What was measured live is the negative half — `GET /api/chapters` answers 200 with chapter 01 present — so the 503 branch is established from the raise sites and the single `writeUnavailable` choke point, and is recorded as that.

---

### 3.6 `GET /api/suggest` — lexical type-ahead. Never embeds.

**Traces:** FR-015, FR-042, SC-005.

**The measured constraint that shapes this endpoint (D-SEARCH-2):** a semantic query embedding was measured at **18.2–21.0 s** under load, plus ~2,202 ms fixed indexer setup per CLI process. The FTS5 prefix index over 58,726 real symbols builds in 0.93 s / 15.9 MB and answers at **p50 0.25 ms, p95 9.58 ms, p99 19.2 ms** — measured under load average ~9 while a full re-index was running. SC-005's 200 ms budget is unreachable semantically and comfortable lexically.

**Normative:**

`/api/suggest` MUST NOT call the embedding backend, MUST NOT call Lumen, and MUST NOT fall back to a semantic path under any condition, including when the lexical index is unavailable. Its `legs` object is always exactly `{"lexical": ...}`. Gate G-HTTP-8 asserts zero embedding requests during suggest traffic.

#### Request

| Parameter | Type   | Default  | Rules                                                                    |
|-----------|--------|----------|--------------------------------------------------------------------------|
| q         | string | required | Trimmed.<br>Empty ⇒ 400<br><code>empty_query</code> . A<br><b>single</b> |

| Parameter | Type   | Default | Rules                                                                                                                                                                                                                                                        |
|-----------|--------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |        |         | <b>character MUST be executed, not short-circuited</b><br>— the ai_interviewing backend’s “return [] below 2 chars” behaviour is a defect and is forbidden here. Longer than 1024 bytes ⇒ 400 query_too_long (a type-ahead query that long is a client bug). |
| limit     | int    | 10      | Max 25.                                                                                                                                                                                                                                                      |
| chapter   | string | —       | Optional scope.                                                                                                                                                                                                                                              |

## Response 200 (envelope of §2.5, payload suggestions)

```

{
  "status": "ok",
  "q": "spec",
  "legs": { "lexical": "ok" },
  "suggestions": [
    { "id": "sg-1", // stable within this
response – needed for aria-activedescendant (FR-042)
      "text": "SpecKit",
      "kind": "symbol", //
"symbol"|"heading"|"chapter"|"speaker"|"term"
      "target": { "pid": "01JBX...", "chapter_slug": "01-ai-
workflows", "t_start_s": 512.4 },
      "score": 12.4 }
    ],
    "total": 1, // for the screen-
reader announcement (FR-041/042)
    "took_ms": 3
  }

```

**Suggestions are navigational targets, not search results** (D-SEARCH-2). Selecting one navigates to a pid; it does not run a semantic query.

**Server-side budget:** `suggest_budget_ms` (default 50, derived/overridable). If the budget elapses with zero rows retrieved, the response is 503 `suggest_timeout` — **not** an empty dropdown. If rows were retrieved, a truncated list is a legitimate ok.

**Unavailable causes:** `lexical_index_unavailable` (FTS5 database missing, locked or corrupt), `index_no_verified_generation`, `suggest_timeout`.

**Accessibility obligations this endpoint carries** (the rest is a frontend contract): stable per-response ids so aria-activedescendant can reference them; a total count so the live region can announce “N suggestions”; deterministic ordering so keyboard traversal is reproducible.

---

### 3.7 GET `/api/search` — on submit

**Traces:** FR-014, FR-016, FR-017, FR-019, FR-020, SC-006, SC-007, SC-008.

#### Request

| Parameter | Type   | Default  | Rules                                                                                                                                                                                                                                                            |
|-----------|--------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| q         | string | required | Trimmed. Empty $\Rightarrow$ 400 <code>empty_query</code> . A multi-paragraph query is legitimate; over 8192 bytes $\Rightarrow$ 413 <code>query_too_long</code> (spec edge case: empty, one-character and multi-paragraph queries must all behave predictably). |
| limit     | int    | 20       | Max 100.                                                                                                                                                                                                                                                         |
| kinds     | csv    | all      | Subset of the kinds the index actually holds — see the note below. The four-value list this cell used to give is <b>WITHDRAWN as measured false</b> .                                                                                                            |
| chapter   | string | —        | Scope to one chapter.                                                                                                                                                                                                                                            |
| mode      | enum   | fused    |                                                                                                                                                                                                                                                                  |

| Parameter  | Type | Default | Rules                                                                                                                                             |
|------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------|
|            |      |         | fused   lexical   semantic.<br>lexical is the honest escape hatch when the operator knows the embedding backend is saturated.                     |
| timeout_ms | int  | 5000    | Clamped to [500, 15000]. This is the bounded timeout D-SEARCH-4 requires; exceeding it ⇒ unavailable{embedding_timeout}, never an unbounded wait. |

**kind, and why pid is NOT always a ULID — measured 2026-09-07, 160 rows over eight queries against the running stack.**

Spec 002 (knowledge-areas-deep-linking) added catalogue kinds to this endpoint and never came back to amend the two clauses below, so §3.7 went on declaring a world that had stopped being true. **A QA challenge then trusted the stale clause**, took `results[0].pid`, built `"/api/passages/" + pid` itself, and reported the resulting 400 `malformed_pid` as a SERVER fault for four separate checks. The server was conforming throughout. This note exists so the next reader does not repeat that.

| kind               | rows | pid shape        | deep_link    |
|--------------------|------|------------------|--------------|
| transcript_segment | 44   | ULID             | supplied     |
| term               | 64   | kg_terms:<ident> | <b>empty</b> |
| doc_section        | 37   | ULID             | <b>empty</b> |
| code               | 14   | ULID             | <b>empty</b> |
| kg_term            | 1    | kg_terms:<ident> | <b>empty</b> |

transcript and diagram — two of the four values the old cell named — did not appear in the sample at all; `transcript_segment` is the value actually emitted.

Two consequences a client MUST honour, both of which the old text implied away:

- pid is a ULID only for kinds that live in the passage registry.** A catalogue row's id is `kg_terms:<ident>`, and §1.4 constrains `{pid}` on `/api/passages/{pid}` to a 26-character ULID — so passing a catalogue id there is a client error, and 400 `malformed_pid` (§5.2) is

the correct answer. **Follow href, which §3.7 gives on every hit, instead of composing a URL.** The catalogue row IS resolvable: GET /api/terms/<ident> returns 200.

2. **deep\_link is empty for every kind but one.** It is the server's statement of where a hit takes the reader, and an empty string means *nowhere* — not *unset*. 116 of the 160 rows carried "" and **0 omitted the key**, so a client using deep\_link ?? fallback never reaches its fallback. Render such a hit as text; do not fabricate a destination from chapter\_slug, which for doc\_section and code rows is docs — not a chapter /api/chapters reports.

## Hit shape

```
{
  "pid": "01JBX7QK3M8V2ZC4YT5N6RWDPA",    // ULID here;
  "kg_terms:<ident>" for a catalogue kind
  "kind": "transcript_segment",              // see the table above
for the live set
  "chapter_slug": "01-ai-workflows",
  "title": "Cutting chunks at silence",
  "snippet": "...never mid-word...",         // built from OUR store;
never upstream free text (§2.4)
  "highlights": [[12, 21]],                 // byte offsets into
snippet; empty for semantic-only hits
  "t_start_s": 512.4, "t_end_s": 519.1,     // transcript kinds
only; null otherwise
  "source_ref": { "path": "chapters/01/transcript/transcript.md",
                  "line_start": 812, "line_end": 815, "symbol":
null },
  "score": 0.71,
  "above_floor": true,
  "leg": "fused",                          // "lexical" |
"semantic" | "fused"
  "provenance": "human_corrected",
  "uncertain": false,
  "href": "/api/passages/01JBX7QK3M8V2ZC4YT5N6RWDPA?context=2",
  "deep_link": "/chapters/01-ai-workflows/
transcript#p-01JBX7QK3M8V2ZC4YT5N6RWDPA"
}
```

**Boundary that must not be implied away** (D-SEARCH-3, FR-031): kind has **no** audio or video value, by design. Lumen's extension allowlist is a compile-time var with no override — 25 code and doc extensions, and **no** image, audio, video, PDF, HTML or SVG. Measured: zero indexed files for .sh .html .css .txt .svg .mmd .puml .vtt .srt .pdf, against 55 tracked .sh, 146 .html and 32 .css in the root module. That is exclusion, not absence. Audio and video are searchable **through their transcripts and captions**; diagrams through their descriptions. FR-016

is satisfied by reduction to text, and the API states the boundary rather than pretending to a capability it does not have. The response carries it in-band:

```
"corpus": {
  "indexed_kinds": ["transcript", "doc_section", "code",
"diagram"],
  "media_reached_via": "transcripts and captions only"
}
```

**Ordering of results — normative. score is NOT the sort key, and a client MUST NOT re-sort by it.**

This clause is new on 2026-09-03. Until today §3.7 stated **no** ordering rule at all, and the served order happened to be descending **score**. It no longer is. **No written contract was violated by that change** — there was nothing here to violate — and that is precisely the argument for writing one: an unstated invariant that clients depend on is a contract the server never agreed to and can break without notice, which is what just happened.

Measured live 2026-09-03 against the running deployment at generation 67, GET /api/search?q=silence&limit=8, ranks 1→8:

```
0.0273  0.0149  0.0143  0.0318  0.0156  0.0149  0.0145  0.0143
               ^^^^^^ rank 4 outscores rank 1
```

Two further queries at the same limit and generation behave the same way: q=chunk reads 0.0164 0.0323 0.0161 ... and q=spec reads 0.0320 0.0307 0.0118 0.0310 .... **The scores themselves did not move; only the order did.**

The ordering that actually applies, in the order it is applied:

1. Candidates are sorted **descending by score**, stably.
2. Hits are withheld and demoted: a hit whose locus does not resolve is dropped entirely (FR-030); a snippet carrying a source-filename shape is dropped by §2.4's disclosure guard; a semantic hit at or below the relevance floor is moved to the near-miss list and is **not** a result. The survivors are truncated to `limit`. **This is the window.**
3. **The window is then replaced by a permutation of itself.** The permutation is a reciprocal-rank fusion ( $k=60$ , the same constant used across legs) of two orderings over the same positions: the incoming score order from step 1, and an Okapi BM25 OR-ranking computed over the window **as its own mini-corpus** — so `idf` and length normalisation are relative to the window, which is what lets a short exact-title row compete with a long transcript segment. The

final sort is stable and tie-broken by incoming position, so two equally-fused documents keep their relative order and an identical request returns an identical order.

4. score is **not** rewritten by step 3. It still means what the leg measured — cosine for semantic, bm25 for lexical, RRF-across-legs for fused. That is why score and position disagree: the number is preserved deliberately, and preserving it is what breaks the correspondence.

**What step 3 cannot do, and these are structural properties rather than promises.** It is a permutation of the served window: it cannot add a hit, cannot remove one, and cannot change a score. So `no_match` (reached from an **empty** hit list) is unreachable through it — a permutation of the empty list is the empty list — the relevance floor is neither read nor widened, and no near-miss can be promoted. It runs on the already-floor-filtered list for exactly that reason.

**The near-miss list is deliberately NOT reordered**, and stays in descending score. It is evidence about what the floor rejected, and its order is the order that produced the rejection.

**This is switchable and the switch is part of the contract.** The permutation is applied when the server's window-rerank option is on. It is on by default and is on in the deployment measured above (the server logs `search window rerank=true` at boot and the running container's argv carries `-search-window-rerank=true`). Turning it off restores step 1's descending-score order exactly, because steps 1 and 2 are unchanged by it. **A client must therefore treat the served order as authoritative and score as a displayable measurement, under either setting** — that rule is the same in both configurations, which is what makes it safe to state.

**Honest boundary (§11.4.6).** Three things above are read from `workshop/platform/backend/pkg/search/` (cited **by path only**) and not from a live probe: the near-miss list's ordering — the three queries measured returned zero near-misses, so no live response exercised it; the determinism of step 3; and the claim that disabling the option restores descending score. On determinism the live evidence is real but thin and is not more than it is: two identical `q=silence&limit=8` requests returned an identical order and identical scores. **Two calls do not establish a determinism property** — they are consistent with it and would have refuted it had they disagreed. The rerank-off

configuration was **not** measured, because doing so requires restarting the running deployment. What was measured live is the fact this clause exists to record: the served order is not descending score.

**SC-006 note:** 2 s p95 is achievable only with a long-lived index child process (avoiding the measured 2.2 s per-process warm-up) **and** embedding capacity reserved from indexing. During a chapter ingest without reservation, 9–50 s was measured. The unavailable state exists for exactly that window, and `reason.retry_after_s` carries the operator's expectation when known.

---

### 3.8 GET `/api/passages/{pid}` — result → exact passage with context

**Traces:** FR-017, FR-037, FR-038, FR-039, SC-003, SC-016.

**Request:** context (int, default 2, max 10) — passages either side.

#### Response 200

```
{
  "passage": { /* the passage object of §3.4 */ },
  "context_before": [ /* up to `context` passages */ ],
  "context_after": [ /* ... */ ],
  "chapter": { "slug": "01-ai-workflows", "title": "..." },
  "recording_seek": { "href": "/api/chapters/01-ai-workflows/recording", "t_start_s": 512.4 },
  "generation": 7
}
```

#### Resolution outcomes — four, and each maps to a distinct HTTP status

| Registry outcome | HTTP            | Body status | Meaning                                                                                    |
|------------------|-----------------|-------------|--------------------------------------------------------------------------------------------|
| found            | 200             | —           | The passage exists and is servable.                                                        |
| redacted         | <b>410 Gone</b> | "redacted"  | The pid is real; its content is suppressed (FR-039). A citation to a redacted passage must |

| Registry outcome | HTTP | Body status   | Meaning                                                                                                                                                                                    |
|------------------|------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| not_in_registry  | 404  | "not_found"   | land on an explicit redaction notice — never on a 404, which reads as a broken link, and never on the content.<br>Determined negative: the registry is readable and this pid is not in it. |
| undetermined     | 503  | "unavailable" | The registry could not be read, or no verified generation is live.                                                                                                                         |
| malformed pid    | 400  | —             | error.code = "malformed_pid".<br>Not a ULID.                                                                                                                                               |

**Normative:** resolution NEVER falls back to fuzzy text matching or nearest-neighbour lookup. An unknown pid is not “probably this one”. Full rules in [passage-contract.md §7](#).

### 3.9 GET /api/passages/{pid}/crossrefs — cross-references

**Traces:** FR-018, SC-016, spec edge case “*cross-references must not create loops that trap a reader*”.

#### Request

| Parameter | Type                                              | Default | Max |
|-----------|---------------------------------------------------|---------|-----|
| relation  | csv of<br>explains,demonstrates,continues,related | all     | —   |

| Parameter | Type                      | Default | Max |
|-----------|---------------------------|---------|-----|
| origin    | derived   authored   both | both    | —   |
| depth     | int                       | 1       | 3   |
| limit     | int                       | 20      | 100 |

**Response 200** (envelope of §2.5, payload crossrefs)

```
{
  "status": "ok",
  "from_pid": "01JBX7QK3M8V2ZC4YT5N6RWDPA",
  "generation": 7,
  "crossrefs": [
    { "to_pid": "01JBX7QM...", "relation": "explains", "origin":
"derived", "score": 0.62,
      "depth": 1, "preview": { "kind": "doc_section", "title":
"...", "snippet": "...", "href": "/api/passages/01JBX7QM..." }
    },
    "truncated_by": null,          // null | "cycle" | "limit" |
"depth"
    "redacted_omitted": 1,        // count only – never the pids,
never the content
    "took_ms": 6
  }
}
```

**Cycle safety (normative):** traversal maintains a visited set; no returned path revisits a pid; from\_pid never appears in crossrefs; self-references are rejected at write time. When traversal stopped because a cycle was detected, truncated\_by: "cycle" says so rather than silently returning a shorter list.

**Redaction:** redacted endpoints are excluded from results, and the *count* is reported. Reporting the count is honest (the reader learns something was suppressed) without leaking what.

**Three states apply:** ok ( $\geq 1$  edge), no\_match (registry read fine, zero neighbours), unavailable (registry/index unreadable, or the derived edges belong to a generation that is not live).

---

### 3.10 POST /api/ask and GET /api/ask/{job\_id} — ask a question, asynchronously

**Traces:** FR-021, FR-022, FR-023, FR-024, FR-025, SC-009, SC-010.

## ROUTE NAMES CORRECTED 2026-09-01 — /api/answer → /api/ask

This section named ~~POST /api/answer~~, ~~GET /api/answer/{job\_id}~~ and ~~GET /api/answering/status~~. Those three names are **withdrawn**, struck through rather than deleted so a reader can tell a corrected name from one that was never wrong. Nothing else in this section changed: every request shape, response shape, status code, normative point A1–A8 and the FR-025 route-separation rule below are unaltered. **Only the spelling of the paths moved.**

- *Believed when written (2026-08-31)*: the contract was authored before the answering tree existed, and /api/answer was chosen on paper with nothing to measure it against.
- *Measured 2026-09-01*, against the live stack on port 8087, and re-derivable in one line each:

```
curl -s -o /dev/null -w '%{http_code}\n' http://
127.0.0.1:8087/api/ask/status # 200
curl -s -o /dev/null -w '%{http_code}\n' 'http://
127.0.0.1:8087/api/ask?q=ping' # 202
curl -s -o /dev/null -w '%{http_code}\n' http://
127.0.0.1:8087/api/ask/nosuchjob # 404
job_not_found
curl -s -o /dev/null -w '%{http_code}\n' -X POST http://
127.0.0.1:8087/api/answer # 404
curl -s -o /dev/null -w '%{http_code}\n' http://
127.0.0.1:8087/api/answering/status # 404
```

platform/backend/pkg/answer/http.go mounts five routes and every one of them is spelled /api/ask. /api/answer and /api/answering/status answer 404 page not found — a plain 404, no contract body.

- *Why the CONTRACT moved and not the code*: /api/ask is the name that exists. It is served by five mounted routes, exercised by pkg/answer/http\_test.go and pkg/answer/citations\_verified\_test.go, probed by platform/gates/route-manifest.tsv, and documented in the workshop's own user-facing pages and training material. /api/answer existed **only on paper**, and its only appearance in executable code was a *negative* assertion — that it must 404. Renaming five working routes would invalidate every measurement taken against them and break documentation that was verified against reality, in exchange for a name whose sole claim is

that this file wrote it down first. A contract earns authority by being the thing the system is measured against; it does not earn it by being older than the system.

- *When it changed:* the answering tree was built at `/api/ask` by a second author as the standalone `cmd/workshop-ask` binary and later folded into `cmd/workshop-server`. The divergence was recorded — never hidden — in `platform/gates/route-manifest.tsv`, which held the three `/api/answer` rows at `NOT_BUILT` specifically so the gap would print as `DEBT` on every run until somebody decided it. This is that decision.
- *What still holds, unchanged:* `POST /api/answer` and `GET /api/answering/status` MUST still answer a **plain 404**, never a 405. A retired name must not start aliasing the live one — two spellings for one resource is the same one-fact-in-two-places defect this correction removes. That assertion did not move into this contract; it already lives in `cmd/workshop-server/web_test.go` (`TestUnimplementedApiRoutesAre404OnEveryVerb`) and `internal/api/gates_test.go`, which is why the manifest rows could be retired without losing the guard.
- *Recorded, NOT resolved by this correction:* the implementation also mounts `GET /api/ask/{job_id}/stream`, which answers (404 `job_not_found` for an unknown id, measured 2026-09-01). **This contract still defers streaming** — see “*Streaming is deferred, not offered*” below, and the verification argument there is untouched by the rename. A route that the contract declines to offer is a **separate, open divergence**, and it is written down here rather than swept in under a renaming decision it has nothing to do with.

## The measurements this contract is built on

- **There is no generative model installed at all.** ollama serves exactly two models and they are the *same* embedding model (`jina-bert-v2`, 160.28M, an encoder with no LM head); the whole model store is 309 MB across 7 blobs — too small to hold a decoder (plan.md premise 5).
- **“Instant” is off by two orders of magnitude.** Three identical two-word embed calls, model resident, minutes apart: **20.16 s / 11.05 s / 0.10 s** at load 8.25. Estimated CPU-only generation (**UNVERIFIED**, §8): **~21 s (1.5B) / ~42 s (3B) / ~95 s (7B)** idle, ×1.4 with verification (D-LLM-5).

- Therefore answering is **asynchronous by contract**, and the extractive adapter — which answers in ~0.3 s, is genuinely grounded and **cannot fabricate** — is the default *working* path.

**POST /api/ask**

### Request

```
{ "question": "why do we cut chunks inside silence?",  
  "chapter": "01-ai-workflows",      // optional scope  
  "top_k": 8 }                       // optional, clamped to the  
configured maximum
```

**Response 202 Accepted** — a job was created:

```
{  
  "job_id": "ans_01JBX7R2...",  
  "poll": "/api/ask/ans_01JBX7R2...",  
  "poll_after_ms": 0,                // 0 for the extractive  
  adapter; ≥2000 for generative  
  "estimated_seconds": null,         // NULL while unmeasured –  
  see §8. Never a guessed number.  
  "provider": { "name": "extractive", "locality": "local",  
  "model": null }  
}
```

**Response 503** — answering cannot run. Returned **immediately**; no job is created:

```
{  
  "status": "unavailable",  
  "reason": {  
    "code": "no_provider",           // §5.3: no_provider |  
    provider_unreachable  
    // |  
    model_not_generative | provider_disabled  
    "message": "Answering is not configured. Search and browsing  
are unaffected.",  
    "retry_after_s": null  
  }  
}
```

model\_not\_generative is a named code because this environment invites exactly that bug: model resolution on this machine lands on an embedding model, and the ollama adapter **must fail construction** with “*model X is an embedding model, not a generative model*” rather than posting /api/chat and rendering the nonsense that comes back.

**Default state:** enabled: false, provider: none. A fresh clone therefore returns 503 here and 200 from /api/search — which is FR-025's acceptance criterion, satisfied out of the box rather than by a runtime check.

**GET /api/ask/{job\_id}**

Exactly four terminal-or-pending shapes:

```
// 200 – still running
{ "status": "pending", "phase": "retrieving",    //
  "retrieving"|"generating"|"verifying"
  "elapsed_s": 12.4, "poll_after_ms": 2000 }

// 200 – answered. Every citation verified against the LIVE
generation's pid set.
{ "status": "answered",
  "text": "Chunks are cut inside measured silence so no word
straddles a boundary...",
  "citations": [
    { "pid": "01JBX7QK3M8V2ZC4YT5N6RWDPA", "quote": "never mid-
word",
      "chapter_slug": "01-ai-workflows", "t_start_s": 512.4,
      "href": "/api/passages/01JBX7QK3M8V2ZC4YT5N6RWDPA" }
  ],
  "retrieval": { "top_score": 0.74, "margin": 0.19, "min_score":
0.55, "min_margin": 0.08 },
  "provider": { "name": "ollama", "model": "...", "locality":
"local" },
  "generation": 7, "elapsed_s": 38.2 }

// 200 – declined. A CORRECT RESULT, not an error.
{ "status": "declined",
  "reason": "margin_too_small",    // §5.5, closed:
below_threshold | margin_too_small
                                   // | unsupported |
no_citations | redacted_evidence
                                   // NEVER an §5.3 code – the
two sets are disjoint (§5.6)
  "message": "The indexed content does not support an answer to
this question.",
  "closest": [ /* Hit[] – what was retrieved, shown as context,
never as an answer */ ],
  "retrieval": { "top_score": 0.61, "margin": 0.02, "min_score":
0.55, "min_margin": 0.08 } }

// 503 – could not determine
{ "status": "unavailable",
  "reason": { "code": "provider_unreachable", "message": "...",
"retry_after_s": 30 } }
```

## Normative points

| #  | Rule                                                                                                                                                                                                                                                                         | Basis                          |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| A1 | declined returns <b>200</b> .<br>Conflating a decline with a failure would make SC-010 unmeasurable — you cannot count refusals if refusals look like outages.                                                                                                               | FR-022, SC-010                 |
| A2 | answered with zero citations is <b>structurally impossible</b> : the generation schema carries "minItems": 1, so an uncited claim is undecodable, and the API rejects any answer object failing it.                                                                          | FR-021, SC-009, D-LLM-2 L2     |
| A3 | Every citation pid MUST be a member of the <b>live</b> generation's pid set — a deterministic set check, and the reason SC-009 is reachable at all. A citation outside the set $\Rightarrow$ the whole answer is declined{no_citations}. Claims are never silently stripped. | SC-009, D-LLM-2 L3             |
| A4 | If a cited passage is redacted between generation and delivery, the answer becomes declined{redacted_evidence}. Redaction propagates to stored answers, not only to the rendered transcript.                                                                                 | FR-039                         |
| A5 | retrieval is returned <b>even on success</b> , so a 0.002-margin pass is visible as <i>fragile</i> rather than indistinguishable from a confident one.                                                                                                                       | data-model<br>Answer.retrieval |

| #  | Rule                                                                                                                                                                                                                                                                                                                                            | Basis                  |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| A6 | min_score/min_margin shipping at 0.0 means <i>uncalibrated</i> , and an uncalibrated threshold <b>refuses everything</b> . “We never calibrated” fails loudly instead of degrading into “answer everything”.                                                                                                                                    | D-LLM-1                |
| A7 | unavailable is a first-class third state, distinct from declined. Conflating them would make a missing provider look like a content gap. Enforced mechanically by §5.6: the decline vocabulary (§5.5) and the answering reason.code vocabulary (§5.3) are <b>disjoint</b> , so no_provider and its siblings cannot be filed as refusal reasons. | FR-025, FR-033, SC-013 |
| A8 | Jobs are in-memory and single-machine. A job id is not durable across a restart; an unknown id ⇒ 404 job_not_found. Claiming durability would be invented capability.                                                                                                                                                                           | D1                     |

**Streaming is deferred, not offered.** A first token can be tens of seconds away, so streaming would materially improve perceived latency — but verification (L3/L4) can only run on a *complete* answer, so streaming would display unverified claims for seconds. If added later, the verified/unverified state must be visible throughout and the answer must not be presentable as final until verification completes.

**GET /api/ask/status**

**Traces:** FR-023, FR-024, FR-031.

```
{
  "enabled": false,
  "provider": "none",
  // none | extractive |
```

```
ollama | openai_compatible
  "model": null,
  "endpoint": "http://127.0.0.1:11434",
  "locality": "local", // DECLARED, never inferred
from the endpoint string
  "locality_verified": true, // resolved-address check
agreed with the declaration
  "calibrated": false, // min_score/min_margin are
still 0.0 ⇒ refuses everything
  "reason": { "code": "no_provider", "message": "No generative
model is installed on this host." }
}
```

**api\_key\_env is never echoed here, never logged, and never included in provider descriptions.** The config holds the *name* of an environment variable, never a value and never a file path.

locality is declared rather than inferred because inferring “127.0.0.1 means local” is a string comparison a typo defeats (localhost.example.com). locality\_verified is the result of resolving the endpoint’s addresses and requiring every one to be a loopback address when locality: local. A divergence between declaration and resolved reality is a **fault**, reported as locality\_verified: false and blocking answering — not a warning.

**FR-025 is enforced architecturally, not by a runtime check:** /api/search, /api/suggest, /api/chapters, /api/passages are registered on a router group with **no reference to the answering package**; /api/ask and /api/ask/status are the only routes that construct a provider. Provider construction failure must not abort server startup. There is no shared goroutine pool or connection pool between the answering client and the search path, so a hung 240 s generation cannot starve search and break SC-006 for a reason unrelated to search. Gate G-HTTP-5 proves it; its paired mutation wires search through the provider health check and must turn the gate red.

---

### 3.11 GET|POST /api/progress — reading position

**Traces:** FR-010.

POST body { "chapter\_slug": "...", "pid": "01JBX...", "t\_seconds": 512.4 }; identity is the X-Session header (the same header ai\_interviewing already uses — FR-013). GET returns the stored position per chapter. Storage is local and single-user (D1); this is a convenience record, not an account.

404 when X-Session has no stored progress (determined negative). 503 when the store is unreadable.

**That 503 carries status: "unavailable" plus reason.code = "progress\_store\_unreadable" and reason.leg = "progress" — §4's ordinary 503 row, with no exception.** Measured in-process on 2026-09-03 by seeding a torn progress.json under an httptest server built from the real handler:

```
// HTTP 503          X-Workshop-Search-Status: unavailable
{
  "status": "unavailable",
  "generation": null,
  "reason": {
    "code": "progress_store_unreadable",    // closed enum, §5.3
    "leg": "progress",
    "message": "the reading-position store could not be read; this
is NOT the same as this session having no stored position",
    "retry_after_s": null                    // §2.5: null is
UNKNOWN, never 0
  }
}
```

Both 503 paths on this route emit that shape — GET when the store cannot be read, POST when the position cannot be stored — from the two non-test raise sites in workshop/platform/backend/internal/api/progress.go (cited **by path only**), each calling the writeUnavailable helper. They share one code because they share one failure: Put reads the store before it writes, so both faults are the same file, and two codes would name one fault twice.

**THE PREVIOUS REVISION OF THIS CLAUSE IS WITHDRAWN, NOT RESTATED — but what it recorded was true when written.** It documented this route as **the one documented exception to §4's 503 row**, because both raise sites called writeError and emitted §1.5's fourth-thing shape instead:

```
// HTTP 503 — PRE-FIX, 2026-09-03. No `status`, no `reason`,
//          and no X-Workshop-Search-Status header at all.
{ "error": { "code": "internal_error", "message": "the reading-
position store could not be read; ..." } }
```

That capture is kept because it is the evidence, and because the deviation is the kind that hides: the property this route actually turns on — that an unreadable store is never reported as “you have never read anything here” — **held throughout**, so the endpoint's own gate

was green the whole time. A 503 was a 503 and a 404 was a 404. What was broken was that this 503 did not *parse* like any other 503, which §6 depends on and no status-code assertion can see.

**Which of the two remedies was taken, and why.** The previous revision named two — teach the route `writeUnavailable` and add a `reason.code` member, or amend §4 to admit an `error.code`-shaped 503 — and made neither. **The first is now applied.** The second was refused: it would spend a contract-wide guarantee, on which every other route and §6's client obligation depend, to accommodate one local convenience store. A new `reason.code` member was required because nothing in §5.3 fits — `registry_unreadable` is `passages.db` and `curriculum_unreadable` is the chapter's pipeline output, and each would send an operator to the wrong file. See §5.3 for the member and §5.6 for its disjointness.

**Gated.** Two tests in `internal/api/progress_test.go` (cited **by path only** — see the note at the end of this clause) carry it: the older one seeds a corrupt store, requires 503, and fails if the body carries a `not_found` verdict; the new one asserts the envelope above on **both** methods — status, an object-shaped reason, membership of `reason.code` in the closed §5.3 enum, the leg, an explicit null `retry_after_s`, and the X-Workshop-Search-Status header. Its paired mutation reverts either `raise site` to `writeError`, which compiles and turns it red. **A client may now assume a 503 from `/api/progress` parses like a 503 from any other route.**

Why the two tests cited in this clause and in §3.7 are named by PATH and never by gate id — the closure check caught this edit, and the catch is worth more than the citation was. The first draft of both clauses cited the implementation's own gate identifiers: the progress-store one here, the search-disclosure one in §3.7. `tasks.md`'s gate-attachment closure check enumerates gate ids by grepping `contracts/` and requires each to be carried by a task line; writing those two into this file moved its population from 31 to 33 and it immediately reported **unattached: 2**, naming both. **The instrument was right.** An id written into `contracts/` is a contract-level obligation that some task must build, and neither of these is one: both name tests that already exist in the backend, cited here as evidence, not commissioned here as work. **The fix was to drop the identifiers — not to widen the check, not to attach them to a task line, and not to touch a single checkbox.** Everything they carried

survives as a path, which is what a reader needs in order to go and look. Re-measured after the correction: **31 ids, unattached: 0.**

**The second half of that lesson is a limitation of the instrument, and it is why this note spells no identifier either.** Because the population is a grep over contracts/, a contract document **cannot discuss a gate id without enlisting it** — including in a note explaining why it should not be enlisted. The first attempt at this very paragraph named both ids and left the check at **33 / unattached: 2**. There is no way to write “this id is deliberately not a contract gate” inside contracts/ and have the check agree. The resolution here is to describe the gates and let the cited test files carry their own names; **deliberately misspelling an id to slip past the extractor was considered and refused**, because an instrument that can be evaded by spelling is worth nothing, and the next reader would have no way to tell a dodge from a typo.

---

### 3.12 GET /api/chapters/{chapter}/accuracy — the measured transcript accuracy

**Traces:** FR-004, FR-033, SC-002.

```
{
  "measured": true,
  "method": "WER against a blind human reference over 30 seeded,
stratified 30-second audio windows",
  "windows": 30,
  "seed": 1724,
  "wer": 0.061,
  "normaliser_sha256": "sha256:...",          // frozen BEFORE the
first measurement
  "companion": { "coverage_gap_rate": 0.0,
"timestamp_error_median_s": 0.4,
                  "timestamp_error_p95_s": 1.9, "speaker_accuracy":
null },
  "evidence_href": ".../_evidence/verify-accuracy/
2026-09-01T09-40Z-.../result.json",
  "measured_at": "2026-09-01T09:40:12Z"
}
```

When the measurement has not been run:

```
{ "measured": false, "wer": null,
  "reason": "verify-accuracy has not been run for this chapter",
```

```
"remedy": "bash workshop/scripts/verify-accuracy.sh 01-ai-  
workflows --reference <path>" }
```

measured: false with wer: null is the third state expressed in data: not “0% error”, not “we failed” — *not measured*. Sampling is on the **audio timeline**, not on machine passages, because sampling passages makes whole-region deletions structurally invisible (a dropped span produces no passage to sample), biasing accuracy upward exactly where the transcript is worst (D-TRANS-3).

---

### 3.13 GET /api/index/status — generation transparency

**Traces:** FR-020, FR-027, FR-033.

```
{  
  "live": { "generation": 7, "state": "live", "pid_count": 1421,  
            "built_at": "2026-09-01T08:02:00Z" },  
  "building": { "generation": 8, "state": "building", "progress":  
0.42 }, // null when idle  
  "legs": { "lexical": "ok", "semantic": "degraded" },  
  "semantic_note": "Upstream index is rebuilding; semantic results  
may be incomplete.",  
  "floor_calibrated": false  
}
```

503 when there is no verified generation at all — a system with nothing live must say so, not serve zero results from an empty index. Only a verified generation may become live, by atomic swap; readers never see building. This is a measured necessity: during the live rebuild, chunks moved 58,734 → 58,744 while last\_indexed\_at/root\_hash still advertised the previous generation — a half-written, readable index (D-SEARCH-5).

Freshness polling (index\_status, measured at 2.6–5.1 s per call) is a **background** poll and MUST NOT be on any request path.

---

## 4. Error and status taxonomy

---

| HTTP | Body discriminator | Category                | Meaning  |
|------|--------------------|-------------------------|----------|
| 200  | status: "ok"       | determined,<br>positive | State 0. |
| 200  |                    |                         |          |

| HTTP | Body discriminator                        | Category                       | Meaning                                                                                          |
|------|-------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------|
|      | status:<br>"no_match" /<br>"declined"     | determined,<br>negative        | State 1. A real,<br>measurable<br>result.                                                        |
| 200  | status:<br>"pending"                      | in progress                    | Answer jobs<br>only.                                                                             |
| 202  | —                                         | accepted                       | Answer job<br>created.                                                                           |
| 206  | —                                         | partial content                | Recording<br>range.                                                                              |
| 400  | error.code                                | request fault                  | Malformed<br>query,<br>malformed<br>pid, empty<br>query. Outside<br>the three states<br>(\$1.5). |
| 404  | status:<br>"not_found" or<br>error.code   | determined,<br>negative        | The thing<br>genuinely does<br>not exist.                                                        |
| 410  | status:<br>"redacted"                     | determined,<br>suppressed      | The pid is real;<br>content<br>withheld<br>(FR-039).                                             |
| 413  | error.code:<br>"query_too_long"           | request fault                  |                                                                                                  |
| 416  | error.code                                | request fault                  | Unsatisfiable<br>or multi-range.                                                                 |
| 503  | status:<br>"unavailable" +<br>reason.code | <b>could not<br/>determine</b> | State 2. Always.                                                                                 |
| 500  | error.code:<br>"internal_error"           | could not<br>determine         | Treated as<br>state 2 by the<br>client; a bug,<br>not a result.                                  |

There is deliberately **no** response in this contract that pairs HTTP 200 with an empty result set and no discriminator. That combination is the failure mode FR-020 exists to prevent.

**The 503 row above has NO exception, and the sentence that used to record one is WITHDRAWN.** An earlier revision named `/api/progress` (§3.11) as the single documented deviation: it answered 503 with an `error.code` of `internal_error` and no `status` and no `reason.code`. That was a correct measurement of the implementation on 2026-09-03, and it has since been **fixed in the backend rather than accommodated here** — both of that route’s raise sites now emit `status: "unavailable"` plus `reason.code = "progress_store_unreadable"` (a new §5.3 member) and `reason.leg = "progress"`, re-measured in-process the same day and gated with a paired mutation. See §3.11 for both captures, before and after.

**The direction of that fix is the point, and it is worth stating once for the next deviation.** The two remedies were to change the code or to widen this row. Widening the row was refused: this row is what lets a client write ONE 503 parser, and §6’s branch-on-`reason.code` obligation is quantified over every route. A contract-wide guarantee is not the right currency for a single endpoint’s convenience store. **A documented exception here is a debt to be paid, not a shape to be matched.**

## 5. Closed enums

**5.1 Inventory.** Six closed enums govern every discriminated field:

| Enum                                             | Members                                                                                               | Where |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------|
| <code>status</code> — search/<br>browse envelope | <code>ok</code>   <code>no_match</code>  <br><code>unavailable</code>                                 | §2.5  |
| <code>status</code> — answer job                 | <code>pending</code>   <code>answered</code>  <br><code>declined</code>  <br><code>unavailable</code> | §3.10 |
| <code>error.code</code>                          | request faults (4xx/<br>5xx)                                                                          | §5.2  |
| <code>reason.code</code>                         | state 2, <i>could not<br/>determine</i>                                                               | §5.3  |
| <code>degraded.*</code>                          | state 0 with a caveat                                                                                 | §5.4  |
| <b>decline reason</b>                            | state 1 negative,<br><i>determined</i>                                                                | §5.5  |

**“Five closed enums” is WITHDRAWN as the count of this table, corrected 2026-09-03 by counting the rows.** The table has six and has had six for as long as this document has existed; the word was never

re-derived after the sixth row was written. The count is the load-bearing part of the sentence — a reader validating a client against “five” has one closed vocabulary they were never told to implement.

**The sixth row was checked for the obvious alternative explanation — that a non-enum had been pasted into an enum table — and it is NOT that.** The decline reason row is a genuine closed enum of the same class as the other five: §5.5 enumerates its members exhaustively, it discriminates a single wire field (reason on a status: “declined” body), and §5.6 makes its disjointness from §5.3 normative rather than advisory. The backend carries it as a distinct closed type with its own validity predicate and its own reason JSON tag, in workshop/platform/backend/pkg/answer/outcome.go (cited **by path only**), where an out-of-set value is rejected rather than serialised. So the row belongs and the number was stale.

**What the miscount actually is a symptom of is visible in the table itself: that row is the only one in bold.** Every other row is plain; this one bolds both its Enum cell and its § cell. Bold on exactly the newest row is the trace of an append that updated the table and not the sentence above it. **Honest boundary (§11.4.6): git cannot separate the two.** `git log -S'Five closed enums'` and `git log -S'decline \reason`over this path both terminate at the same single commit695c22d``, so the ordering above is read from the formatting, and the formatting is evidence of a late edit, not proof of one. It is recorded as the reading it is.

status carries two vocabularies because it discriminates two different resources; both are closed, and the resource decides which applies. Closed means a value outside the enum is a contract violation, not an extension point. A new failure mode gets a new enum member and a new gate — never a free-text string.

**5.2 error.code (4xx/5xx request faults)** empty\_query, query\_too\_long, malformed\_pid, unknown\_parameter, invalid\_range, multi\_range\_unsupported, chapter\_not\_found, material\_not\_found, job\_not\_found, internal\_error, transcript\_not\_produced, area\_not\_found, area\_not\_published, term\_withdrawn, term\_not\_found.

**The last five were added on 2026-09-03 by measurement, not by design, and — as in §5.3 — the direction of the gap was implementation-ahead-of-contract, not the reverse.** All five are emitted on the wire today by a deployment this contract governs; none of them was listed here. §5.1’s rule cuts the same way it does for reason.code: while a row was missing, a conforming client validating

against the printed contract had to reject a response the server legitimately produced. Every raise site is in workshop/platform/backend/internal/api/ and is cited **by path only**.

| Code                    | HTTP | Raised when                                                                                                                                                                                                                                              | Evidence it reaches the wire                                                                                                                                                                                                        |
|-------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| transcript_not_produced | 404  | §3.4’s SECOND determined negative: the chapter <b>exists</b> and nothing has transcribed it. Deliberately not chapter_not_found — “no such chapter” sends a reader to the chapter list, “not transcribed yet” sends them to the pipeline.                | Non-test raise site in internal/... asserted on the response body chapters_test.go, which requires error.code == "transcript_not_found" states in its own failure message chapter_not_found is the wrong test ./internal/api/ PASS, |
| area_not_found          | 404  | The area id names no area at all — either absent from the taxonomy, or 002 A3.1.1 applies (zero live evidencing mentions, which that contract treats as never having been returned).                                                                     | Live probe 2026-09-03 against deployment: GET /api/areas/00000000000000000000000000000000 {"error": {"code": "area_not_found", "area", "field": "area"}}. Test two each in areas.go, question coverage.go and export.go.            |
| area_not_published      | 404  | 002 A3.2.2: the area is <b>real</b> and has live evidence, but carries no recorded publication review, and A3.2.2 says plainly that the area is then not served. Distinct from area_not_found — what is missing is a different fact than “no such area”. | Live probe 2026-09-03: GET 01M1GWQYWSKZA7A2S59MNV0Z20 GET /api/areas on that same 404, body {"error": {"code": "area_not_published", area has no recorded publication FR-016/FR-048 make a review serve an area's materials'        |
| term_withdrawn          | 404  | 002 A3.4.2: the term is withdrawn from the taxonomy rather than                                                                                                                                                                                          | Two non-test raise sites in internal/terms.go; asserted on the response by internal/api/terms_test.go                                                                                                                               |

| Code           | HTTP | Raised when                                                                                                                                                                                                                                                                            | Evidence it reaches the wire                                                                                                                                                                                             |
|----------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                |      | merely unlinked. Two causes, one code, each naming its own reason in the message:<br>all_evidence_redacted (evidence existed at build time and was redacted or removed) and<br>no_evidence_at_build (the taxonomy build never recorded any — an FR-008 build defect, not a redaction). | error.code == "term_withdrawn" internal/api/ PASS, 2026-09-03<br><b>provable on this deployment</b> reports withdrawn: {"count": 8537, "term": "term"} over 8,537 terms at generation time. term is in that state today. |
| term_not_found | 404  | The taxonomy never carried a term of that name at all — as opposed to one withdrawn (above).                                                                                                                                                                                           | Live probe 2026-09-03: GET knowledge-route-manifest-p {"error": {"code": "term_not_found", "field": "term"}}.                                                                                                            |

### transcript\_not\_produced closes an inconsistency the implementation had already recorded against this document.

internal/api/api.go's declaration of that constant carries an in-source HONEST BOUNDARY (§11.4.6) note saying §3.4's prose names the code while §5.2's printed inventory does not, and resolving it in favour of the endpoint clause that actually describes the response. That note was correct and is now discharged from this side: §3.4 and §5.2 agree.

### Honest boundary (§11.4.6) — what these five rows do NOT claim.

They are a statement about the codes named here, established one at a time; they are not a re-audit of §5.2 as a whole. job\_not\_found and invalid\_range were carried by this list before today and were **not** re-verified against the wire on 2026-09-03, so nothing above should be read as evidence that the remaining ten members are each implemented. term\_withdrawn in particular is established from its raise sites and its passing wire-level test, **not** from a live response — the running deployment holds no withdrawn term to produce one.

### 5.3 reason.code (state 2 — could not determine)

| Code                         | Leg        | Raised when                                                                                                                                                                                                                                                                                          |
|------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| embedding_timeout            | semantic   | timeout_ms elapsed on the query embed. The direct fix for the recorded unbounded stall.                                                                                                                                                                                                              |
| embedding_backend_exhausted  | semantic   | Upstream reported all embedding servers exhausted.                                                                                                                                                                                                                                                   |
| embedding_degenerate_vectors | semantic   | Background distinct-vector process failed — the mode that returns HTTP 200 with a repeated stale vector: well-formed, non-NaN, correct L2 norm, and completely wrong. It put 758 duplicate vectors into this index on 2026-08-26.                                                                    |
| partial_failure_zero_results | mixed      | Invariant I5: a leg failed and the survivors found nothing.                                                                                                                                                                                                                                          |
| lexical_index_unavailable    | lexical    | FTS5 database missing, locked or corrupt.                                                                                                                                                                                                                                                            |
| code_index_unavailable       | lumen      | The code-semantics leg's own backend could not be reached. A separate member from lexical_index_unavailable, not reuse of it: a client that retries, an operator who reports, has to tell <b>which</b> backend went down and borrowing the other leg's code sends both to the wrong one.             |
| index_no_verified_generation | index      | Nothing is live.                                                                                                                                                                                                                                                                                     |
| index_rebuilding_no_fallback | index      | Upstream rebuilding <b>and</b> the lexical leg is also down.                                                                                                                                                                                                                                         |
| suggest_timeout              | lexical    | suggest_budget_ms elapsed with zero rows.                                                                                                                                                                                                                                                            |
| registry_unreadable          | registry   | passages.db cannot be opened.                                                                                                                                                                                                                                                                        |
| curriculum_unreadable        | curriculum | The chapter's own committed pipeline output — the transcription sidecar carrying the engine, the duration and the segment spans could not be read (§3.3, §3.4). Not reuse of registry_unreadable, because the two remedies point to different files: an unreadable registry is fixed by re-ingesting |

| Code                                                                            | Leg       | Raised when                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                 |           | unreadable curriculum tree by repairing the mount or re-running transcription.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| progress_store_unreadable                                                       | progress  | <p>The local reading-position store (progress.json) exists and could be used (§3.11), on either the GET read path or the POST write path. Put reads before it writes, so both are the same file and the same fault. Not a reuse of registry_unreadable or curriculum_unreadable: this is a <b>third</b> file with a <b>third</b> remedy: repair or delete one local convenience file, costing a read of their saved positions and nothing else — and borrowing either existing code would send an operator to the wrong one. progress is likewise a leg of its own and never appears in a search envelope’s legs map, because § is not a retrieval route.</p> |
| recording_not_reassembled                                                       | —         | 36 parts present, no reassembly file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| recording_integrity_unverified/<br>_stale/_mismatch                             | —         | FR-007 states.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| too_many_readers                                                                | —         | Concurrent range readers exceeded.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| no_provider/provider_disabled/<br>provider_unreachable/<br>model_not_generative | answering | FR-025 states.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| verification_unavailable                                                        | answering | <p>Citation verification itself could run — never treated as “verification”. Names the <b>passage-facing</b> layer (L3 citation identity, L4 support the fix, when there is one, is on passage side.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| question_verification_unavailable                                               | answering | <p>The <b>answer-against-question</b> (L5) was configured, was consistent and <b>could not decide</b> — the m</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Code                                                | Leg       | Raised when                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     |           | <p>endpoint went away, the deadline elapsed, or the question's domain could not be classified.</p> <p><code>min_score / min_margin</code> are still too low so the admission gate cannot separate answerable from unanswerable. It refuses <b>as standard</b> never as a decline: an uncalibrated instrument has judged nothing, so it is filing its refusal under §5.5 without blame the corpus for a missing calibration (A6, D-LLM-1, §8 US).</p>             |
| <code>thresholds_uncalibrated</code>                | answering |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>ingest_in_progress</code>                     | answering | <p>An exclusive ingest lock is held, so answering is suspended while the search keeps serving the existing live generation (D-LLM-10).</p>                                                                                                                                                                                                                                                                                                                       |
| <code>request_cancelled</code>                      | answering | <p>The caller went away, or the deadline elapsed, before any verdict existed. No verdict was reached, so none may be reported.</p>                                                                                                                                                                                                                                                                                                                               |
| <code>locality_unverified</code>                    | answering | <p>The declared <code>locality</code> disagreed with the endpoint's resolved addresses (§3.10 GET <code>/api/ask/...</code> status). A fault that blocks answering outright — never a warning served alongside an answer.</p>                                                                                                                                                                                                                                    |
| <code>generation_gated_pending_clarification</code> | answering | <p>Retrieval admitted material flaws as requiring answer-against-question verification while that layer is <b>not built in this deployment</b>. Distinct from <code>verification_unavailable</code>, which is transient and may succeed on next call, and from <code>question_verification_unavailable</code> which names a layer that existed but did not decide: this one is <b>standalone</b> and no restart, provider change or configuration clears it.</p> |

**An EIGHTH row, `progress_store_unreadable`, was added on 2026-09-03 for the opposite reason to the seven below — it is the only member here that the implementation did NOT already emit.** It was minted to close §4's one documented 503 deviation (§3.11), and it was minted only after the existing members were checked against the fault and found to misdescribe it. It is therefore *not* a counter-example to the rule stated below: the code and the contract row landed **together**, in the same change, with a paired-mutation gate, so there was never an interval in which a conforming client would have rejected a legitimate response.

**Seven rows above were added on 2026-09-03 by measurement, not by design, and the direction of the gap was the opposite of the one assumed.** They are `code_index_unavailable`, `curriculum_unreadable`, `thresholds_uncalibrated`, `ingest_in_progress`, `request_cancelled`, `locality_unverified` and `generation_gated_pending_clarification` — seven codes that the implementation **already emits on the wire** and that this printed table did not list. The enumeration was taken from the two closed vocabularies the backend declares against this section by name, in `workshop/platform/backend/pkg/search/envelope.go` and `workshop/platform/backend/pkg/answer/outcome.go`, each cited **by path only**; every one of the seven has at least one non-test raise site. Two of them carry an in-source note saying this table lacks the row and asking for it.

**This matters because §5.3 is closed.** “A value outside the enum is a contract violation, not an extension point” (§5.1) cuts both ways: while the row was missing, a conforming client validating against the printed contract would have had to reject a response the server legitimately produced. A code that is implemented therefore belongs here **more** urgently than one that is not — the reverse reading, that this section is a register of unbuilt codes, is **withdrawn**; `registry_unreadable`, `suggest_timeout` and `embedding_timeout` have all been implemented since before it was written. Codes that are contracted and unbuilt are tracked in §8 and in `tasks.md`, never by omission here.

**`question_verification_unavailable` is a third code, not a reuse of `verification_unavailable`, and the difference is operational.** `verification_unavailable` says the machinery that checks a claim *against its passages* did not run. This one says the machinery that checks a claim *against the question* did not run. An operator reading them needs to know which side to look at, and one code covering both would not tell them.

**It is never a degrade.** Where a layer was asked for and then produced no verdict, its verdict may not be supplied by some other layer that happens to have completed ([specs/002-knowledge-areas-deep-linking C4.3.4, FR-052](#)). Serving answered on the strength of the four passage-facing layers, because the question-facing one was silent, is exactly the substitution that requirement forbids. This code is what refusing that substitution looks like on the wire: state 2, 503, and no answer.

**5.4 degraded.\* (state 0 with a caveat)** semantic: reindexing | partial | stale\_generation. lexical: stale\_generation.

### 5.5 decline reason (state 1 — determined, negative)

Carried as a **bare enum string** on status: "declined" bodies (§3.10), never as an object. This is the only registry of these members; a document that lists them elsewhere is quoting, not defining.

| Member            | Raised when                                                                                                                                                        |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| below_threshold   | Top retrieval score is under min_score. The model was never called (L1).                                                                                           |
| margin_too_small  | Top-1 to top-2 margin is under min_margin — retrieval could not discriminate.                                                                                      |
| unsupported       | Support verification (L4) found a claim the cited passages do not entail.                                                                                          |
| no_citations      | A citation pid is outside the live generation's member set (A3), so the whole answer is declined rather than silently stripped.                                    |
| redacted_evidence | A cited passage was redacted between generation and delivery (A4).                                                                                                 |
| does_not_answer   | Answer-against-question verification (L5) found a claim that its cited passages <b>do</b> support and that answers a <b>different question</b> than the one asked. |

**Why does\_not\_answer is not unsupported, and may never be folded into it — normative.** The two sound adjacent and are opposites. unsupported means *the cited passages do not state the claim*: the

generator produced content the corpus does not carry. `does_not_answer` means *the citation is exact and the claim is still off-target* — typically a fragment lifted out of a passage on a related topic, which clears every passage-facing layer by construction, and a verbatim fragment clears them by identity.

**Their remedies point in opposite directions.** An unsupported claim needs better evidence — the answer is about the right thing and rests on nothing. A `does_not_answer` claim needs a different answer — the evidence is impeccable and the answer is about the wrong thing. Reporting the second under the first would tell a maintainer to go fix retrieval when the retrieval was fine, and it would make the two rates indistinguishable in aggregate, so a system trading one failure for the other would look unchanged. Reporting it under the first was considered and **refused** for exactly that reason; the member exists so the distinction survives to the wire.

This is the layer that closes the gap the earlier four could not see: [specs/002-knowledge-areas-deep-linking](#) C4.3.3 (FR-051) records that the four existing layers all verify the claim against the **passage** and none verifies it against the **question**, which is how a topically related fragment passes all four. A claim that fails L5 refuses the **whole** answer — no claim is stripped and the remainder served — on the same rule as `no_citations` above.

## 5.6 The two answering vocabularies are disjoint — normative

§5.5 and the answering-leg rows of §5.3 (`no_provider`, `provider_disabled`, `provider_unreachable`, `model_not_generative`, `verification_unavailable`, `question_verification_unavailable`, `thresholds_uncalibrated`, `ingest_in_progress`, `request_cancelled`, `locality_unverified`, `generation_gated_pending_clarification`) **share no member, and no member may be added to both.** A decline is a judgement about the *content*; unavailable is the absence of the instrument that would have made that judgement. Filing `no_provider` as a decline reason would report a thing that could not run as a thing that was judged — state 2 rendered as state 1, which is the precise failure the three-state contract exists to prevent. This rule is what makes A7 checkable rather than merely stated: the disjointness is mechanical, so a violation is a schema error and not a matter of interpretation.

**`progress_store_unreadable` is disjoint from §5.5 too, and is deliberately NOT added to the enumeration above.** That list is scoped, by its own opening words, to the **answering-leg** rows of §5.3,

and this member's leg is progress — appending it would misstate the section's own scope and quietly imply `/api/progress` has an answering path, which it does not. The disjointness it needs is satisfied on stronger grounds than membership of a list: §5.5's vocabulary describes a judgement about *content* reached by the answering pipeline, and §3.11 neither judges content nor has a pipeline, so no decline reason can name this fault and this code can never name a decline. Recorded here rather than left inferred, because “it is not in the list” and “it is not disjoint” are different facts and only the first is true.

---

## 6. What the client is contractually required to do

The API cannot enforce rendering, so these obligations are stated here and tested by G-HTTP-7.

1. **Three render branches, minimum**, keyed on `body.status` — never on `results.length`.
  2. **Never render the no-results copy on a non-200 response**, and never on `status != "no_match"`.
  3. `reason.evidence` and `degraded.evidence` are diagnostics; render them only inside an explicit disclosure, never as primary copy.
  4. `near_misses` render in a distinct, labelled region: *below the relevance threshold*.
  5. When `floor_calibrated == false`, a `no_match` renders with the calibration caveat (I7).
  6. When `degraded` is present, a persistent banner is shown for the whole result set — not a dismissible toast that leaves the reader believing they saw everything.
  7. Suggestions are announced through a live region using `total`, and are keyboard-traversable via the per-response `ids` (FR-041, FR-042).
  8. `410 redacted` renders as an explicit redaction notice, never as a broken link.
-

## 7. Traceability

### 7.1 Requirements this contract covers

| Requirement                                   | Where                                                                 |
|-----------------------------------------------|-----------------------------------------------------------------------|
| FR-001 complete transcript                    | §3.4 coverage block                                                   |
| FR-002 timestamps                             | §3.4 t_start_s/t_end_s; §3.5.2 seek                                   |
| FR-003 uncertainty marked                     | §3.4 uncertain, uncertain_reason                                      |
| FR-004 accuracy report                        | §3.12                                                                 |
| FR-005 speaker attribution / stated inability | §3.4 speaker, speaker_source                                          |
| FR-007 integrity verified, fail loudly        | §3.5.2 integrity table                                                |
| FR-008 browsable chapters                     | §3.2                                                                  |
| FR-009 read + play in place                   | §3.3, §3.4, §3.5, §3.5.3                                              |
| FR-010 progress preserved                     | §3.11                                                                 |
| FR-011 navigable sections                     | §3.3 sections, §3.4 section filter                                    |
| FR-013 conventions of the reference module    | §1.1 base path, X-Session in §3.11                                    |
| FR-014 meaning-based results                  | §3.7 mode=fused/semantic                                              |
| FR-015 suggestions while typing               | §3.6                                                                  |
| FR-016 all content types indexed              | §3.7 kinds + corpus boundary                                          |
| FR-017 result → exact passage with context    | §3.8                                                                  |
| FR-018 cross-references                       | §3.9                                                                  |
| FR-019 plain no-match, no unrelated answers   | §2.6 I1/I3/I6                                                         |
| FR-020 honest unavailability                  | §2 in full; §5.3                                                      |
| FR-021 answers with citations                 | §3.10 A2, A3                                                          |
| FR-022 decline rather than fabricate          | §3.10 declined, A1                                                    |
| FR-023 local and external providers           | §3.10 /api/ask/status                                                 |
| FR-024 no external transmission when local    | §3.10 locality, locality_verified<br>§3.10 route separation; G-HTTP-5 |

| Requirement                                 | Where                                                                      |
|---------------------------------------------|----------------------------------------------------------------------------|
| FR-025 search survives answering outage     |                                                                            |
| FR-027 idempotent extension (observable)    | §1.2 X-Workshop-Index-Generation; §3.13                                    |
| FR-031 honest statement of limits           | §3.7 corpus; §3.10 reason; §3.12 measured:false                            |
| FR-032 every check has a paired mutation    | §2.7                                                                       |
| FR-033 unable-to-verify distinguishable     | §2.1, §4, §3.12, §3.10 A7                                                  |
| FR-037 stable identifiers                   | §1.4, §3.8 (contract in passage-contract.md)                               |
| FR-038 immutable machine layer + provenance | §3.4 provenance, machine_text                                              |
| FR-039 redaction propagates                 | §3.4 markers, §3.8 410, §3.9 redacted_omitted, §3.10 A4                    |
| FR-041 / FR-042 accessibility               | §3.6 id/total/ordering; §6 client obligations                              |
| SC-001 100% coverage accounted              | §3.4 coverage.unexplained_gap_s                                            |
| SC-002 measured accuracy published          | §3.12                                                                      |
| SC-003 land within 5 s                      | §3.4 timestamps + §3.5.2 range support                                     |
| SC-005 suggestions $\leq 200$ ms p95        | §3.6 lexical-only, suggest_budget_ms                                       |
| SC-006 results $\leq 2$ s p95               | §3.7 timeout_ms; §3.13 generation transparency                             |
| SC-007 / SC-008 retrieval quality           | §3.7 mode, score, above_floor — <b>quality itself is unverified, §8 U5</b> |
| SC-009 citations resolve and support        | §3.10 A2/A3                                                                |
| SC-010 declines, never fabricates           | §3.10 A1/A6                                                                |
| SC-013 three states in 100% of checks       | §2.1, §4                                                                   |
| SC-016 references survive correction        | §3.8 resolution table; contract in passage-contract.md                     |

| Requirement               | Where                           |
|---------------------------|---------------------------------|
| SC-017 WCAG AA + keyboard | §3.6, §6 (API obligations only) |

## 7.2 Requirements not covered by this contract, and where they live

| Requirement                                         | Owner                                                                                                                                                  |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-006 preserve source material unmodified          | <a href="#">pipeline-cli.md §4.1</a> — read-only guarantee at ingest                                                                                   |
| FR-012 documented start/stop                        | <a href="#">pipeline-cli.md §4.8</a>                                                                                                                   |
| FR-026 / FR-028 / FR-029 extension procedure        | <a href="#">pipeline-cli.md §4.7, §4.1</a>                                                                                                             |
| FR-030 documentation set                            | Not a contract. docs/ deliverable + quickstart.md.                                                                                                     |
| FR-034 no server-side CI                            | <a href="#">pipeline-cli.md §2.7</a>                                                                                                                   |
| <b>FR-035 governance carriers in lockstep</b>       | <b>No interface contract can cover this.</b> It is a repository governance task (scripts/verify-governance-cascade.sh). Flagged, not silently dropped. |
| <b>FR-036 clean tree across repo and submodules</b> | <b>No interface contract can cover this.</b> Operator/commit-wrapper task. Flagged.                                                                    |
| FR-040 evidence to a versioned location             | <a href="#">pipeline-cli.md §2.5</a>                                                                                                                   |
| FR-041 / FR-042 full WCAG conformance               | Frontend contract + SC-017 audit. This document specifies only the API affordances that make them attainable.                                          |

## 8. Unverified register — what this contract depends on that has not been measured

Marked so no reader mistakes a design intent for a measurement. Each names what settles it.

| #                                     | Depends on                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Status     | Settled by                                                                                                                                                                                                                                                                                                           |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| U4<br>(research.md)                   | <p>Idle embedding latency. /api/search's SC-006 budget and the default timeout_ms: 5000 assume the semantic leg can answer in seconds when idle. Only loaded figures exist (18.2–21.0 s, and 20.16/11.05/0.10 s across three identical calls).</p> <p>Whether the ~2.2 s per-process indexer setup amortises across calls in one long-lived session. If it does not, every semantic query carries a multi-second fixed cost and SC-006 is unreachable regardless of U4.</p> | UNVERIFIED | Time ten consecutive semantic_search calls in one warm session against an idle backend, <b>after</b> the running lumen index --force rebuild completes.                                                                                                                                                              |
| C-3 (research/search-architecture.md) | <p>The relevance floor separating ok from no_match. No corpus exists yet, so no floor has been calibrated.</p>                                                                                                                                                                                                                                                                                                                                                              | UNVERIFIED | <p>Same ten-call timing run; inspect the spread, not just the mean.</p> <p>Build the SC-007/SC-008 benchmark set (<math>\geq 20</math> labelled queries) after Chapter 1 is indexed; calibrate, then flip floor_calibrated to true. Until then invariant <b>I7</b> applies and the API reports floor_calibrated:</p> |
| U5<br>(research.md)                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | UNVERIFIED |                                                                                                                                                                                                                                                                                                                      |

| #                          | Depends on                                                                                                                                                                                          | Status            | Settled by                                                                                                                                                                    |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            |                                                                                                                                                                                                     |                   | false on every response.                                                                                                                                                      |
| <b>D-LLM-5 estimate</b>    | Generation latency (~21 s / ~42 s / ~95 s idle by model size). No generative model is installed, so nothing was timed.                                                                              | <b>UNVERIFIED</b> | An operator ollama pull of a generative model, then a timed run. <b>Until then estimated_seconds MUST be null</b> — the API does not publish an estimate it has not measured. |
| <b>HelixLLM behaviour</b>  | The openai_compatible adapter is claimed to cover HelixLLM. No HelixLLM server is running and no checkout exists on this disk; nothing about it was tested.                                         | <b>UNVERIFIED</b> | Start a HelixLLM instance and run the adapter's conformance check against it.                                                                                                 |
| <b>U3</b><br>(research.md) | That the recording contains intelligible speech in a known language. Nobody has listened and no ASR has run. Every transcript-serving endpoint here presumes a transcript exists and is meaningful. | <b>UNVERIFIED</b> | The 5-minute calibration run on the already-extracted 300 s sample (also closes U1 and U2).                                                                                   |

**Not marked unverified, because they were measured:** FTS5 prefix latency (p95 9.58 ms), health\_check lying about saturation, Lumen's glued no-results/warning string, Lumen's content-derived chunk ids,

the half-written readable index during rebuild, the recording's duration and dual-mono audio, and the `search.component.ts` anti-pattern.